Economic Regime Radar:

"Hybrid Forecasting Meets Regime Classification — One Quarter at a Time"

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What is an "Economic Regime"?

Regime	What it means	Strategic Implication
Boom	Strong growth, lots of jobs, Industrial Acceleration	Expand, hire, invest
Stability	Calm growth, steady signals	Stay the course
Slowdown	Economy starting to weaken	Be cautious, Manage Risk
Recession	Economy Shrinking, Demand Falling	Cut Costs, Hold Back

The Problem this Solves:

"Can't make the right decision without really knowing what's happening next Quarter!"

Stakeholders: Financial leaders like CFOs, hedge fund managers, and central banks need clear, forward-looking signals to guide budgeting, hiring, monetary decisions, and investment.

Problem: Traditional economic forecasts are considered vague and binary — offering little detail on what's coming next, and too late to act on.

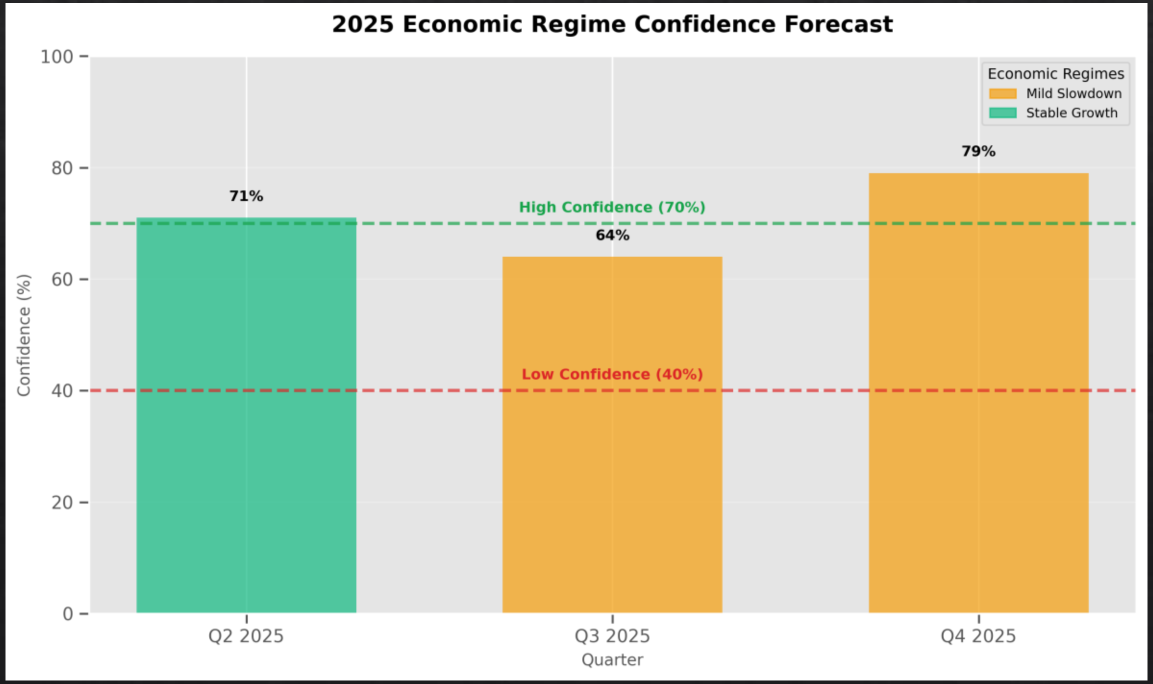
Objective: Build a hybrid time series model that forecasts key macroeconomic indicators and classifies upcoming quarters into clear, interpretable economic regimes — helping leaders make decisions before the shift.

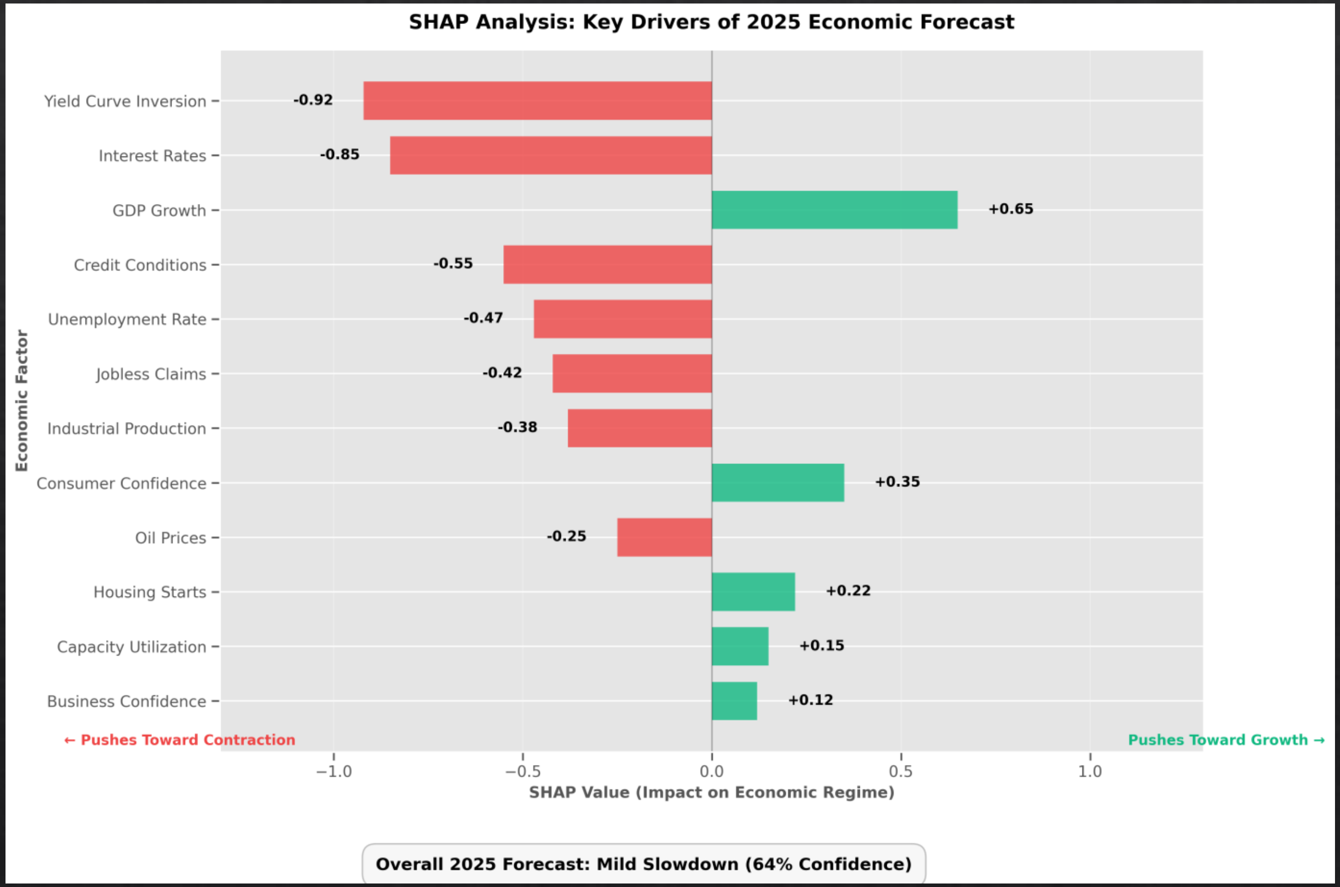
The Big 3 Steps we took:

Look	Look at key signals like GDP, unemployment, & such
Predict	Predict what economic regime is next (XGBoost Classifier)
Interpret	Interpret what really caused the Economic shift (SHAP)

Upcoming Results for 2025

Quarters	Economic Regimes	Key Contributing Indicators
2025Q2	Stable Growth (71%)	GDP Growth + , Business Sentiment , Industrial Production +
2025Q3	Mild Slowdown (64%)	GDP Growth +, Interest Rates -, Yield Curve Inversion +
2025Q4	Mild Slowdown (79%)	GDP Growth +, Unemployment Rate + Yield Curve Inversion +



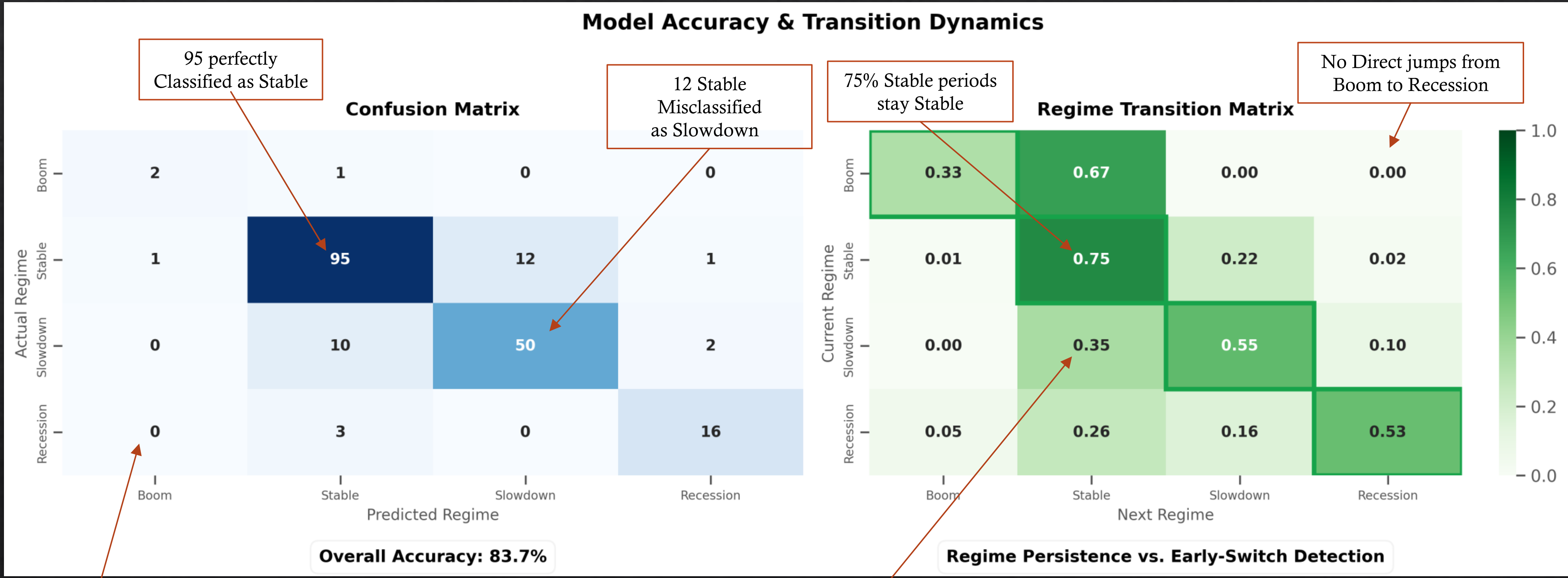


Model Choice — Why We Used XGBoost Classifier

- Had most Accurate Testing Scores for each regime
- Helped best with balancing Messy, Non-Linear, Disorganized Data
- Worked considerably well with SHAP
- Allow you to extract probability scores best

	precision	recall	F1-score	support
Boom	0.833	1.000	0.909	5
Recession	0.875	1.000	0.933	7
Stable	0.909	0.833	0.870	24
Slowdown	0.692	0.692	0.692	13
accuracy	0.837	0.837	0.837	

Matrices Results using XGBoost Classifier:



No Misclassification between "Boom" & "Recession"

Frequent rebounds from Slowdown to Stable (35%)

How we built the Model:

Example of Data Cleaning (SQL):

```
# --------------------------------------------------------
# Inventories-to-Sales Ratio (with division-by-zero guard)
conn = sqlite3.connect(':memory:')
Business_Inventories_df.to_sql('Inventories', conn, index=False, if_exists='replace')
Retail_Sales_df.to_sql('Sales', conn, index=False, if_exists='replace')

sql_query = """
SELECT Sales.observation_date, Sales.Total_Retail_Sales,
Inventories.Business_Inventories
FROM Sales
LEFT JOIN Inventories
ON Sales.observation_date = Inventories.observation_date
WHERE Sales.observation_date >= '1992-01-01'
"""

Inventories_to_Sales_Ratio_df = pd.read_sql_query(sql_query, conn)
Inventories_to_Sales_Ratio_df['Inventories_to_Sales_Ratio'] = np.where(
    Inventories_to_Sales_Ratio_df['Total_Retail_Sales'] == 0,
    0,
    Inventories_to_Sales_Ratio_df['Business_Inventories']
    / Inventories_to_Sales_Ratio_df['Total_Retail_Sales']
)
conn.close()
# --------------------------------------------------------
# Volatility Shock
Volatility_Index_df['Volatility_Index_Growth_Rate'] = Volatility_Index_df[
    'Volatility_Index'
].pct_change() * 100
```

Data Sources: FRED, BLS
Libraries: ‘Pandas’, ‘sklearn.metrics’, ‘SQLite’, ‘Statsmodel’, etc.

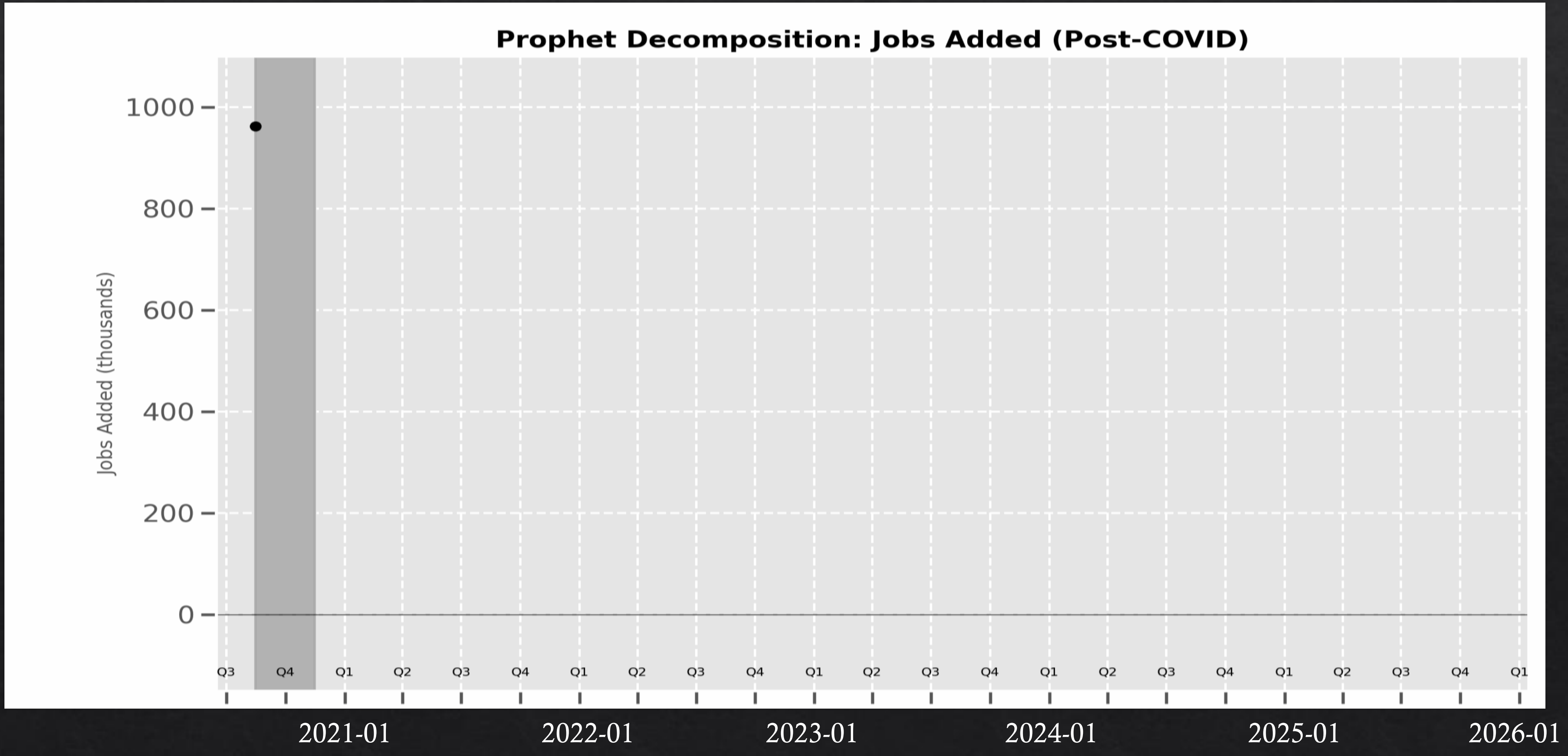
Modeling Steps:

- ◇ Customized over 30 key Economic signals
- ◇ Fix Data imbalance through smart sampling
- ◇ Forecasted upcoming Indicator Results
- ◇ Trained a Model to recognize Economic Patterns based on Past Trends

Feature Engineering *(Creation of 304 new Economic Indicators)*:

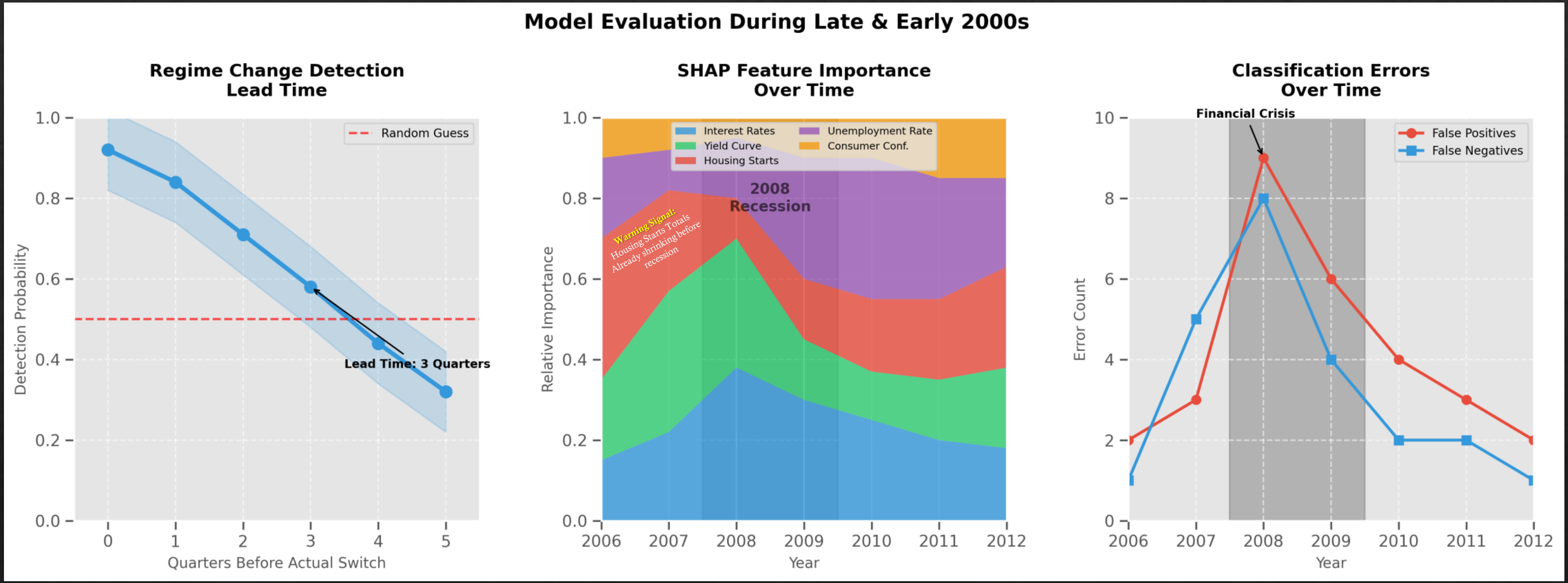
- ◇ Looked at Recent History for Each Signal
- ◇ Captured Patterns like Averages, % Changes, & Standard Deviations
- ◇ Built Custom Complex Indicators

Example: *Prophet Decomposition (Post-Covid):*



Evaluation Highlights:

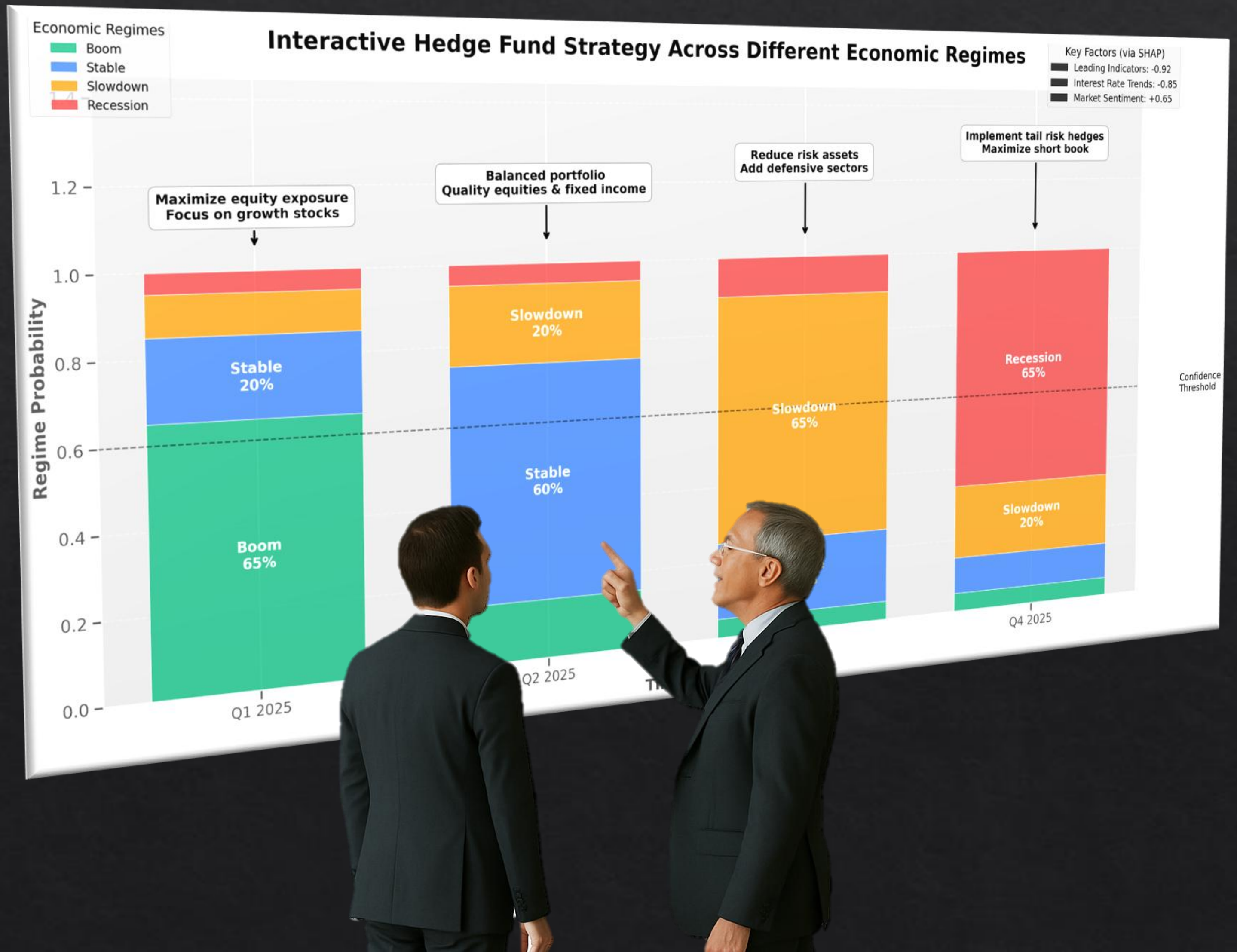
"Catching Early Signals before 2008 Downturns"



Business Recommendations

Recommendation for each Stakeholder:

- **Chief Financial Officers:** Adjust budgets and hiring based on forecast regime
- **Hedge Fund Managers:** Rebalance portfolios before regime switch
- **Central Banks:** Use early signals to guide monetary response



What's Next?

More than just Numbers, use data incorporate Behavioral Signals from News & Sentiment too to predict

Shift from Quarterly to Monthly Forecasting

Build an Interactive Tool for Custom Forecasts

Summary

What This Project Did:

- Looked ahead at things like GDP, jobs, and inflation
- Predicted where the economy is headed next — Boom, Stable, Slowdown, or Recession?
- Explained **why** those changes are coming

Why It Matters:

- Helps business and finance leaders prepare early
- Clear reasons behind each prediction
- 84% accurate in spotting past changes — before they happened

ANY QUESTIONS?